





[illegible]

|           |  |  |
|-----------|--|--|
| Variables |  |  |
|-----------|--|--|

[illegible]

**Constraints**

Permutation Constraints

Marginals Fixed

|             | Monge                   | Kantorovich     |
|-------------|-------------------------|-----------------|
| Variables   |                         |                 |
| Constraints | Permutation Constraints | Marginals Fixed |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |

| Type |  |  |
|------|--|--|
|------|--|--|

|             | Monge                   | Kantorovich     |
|-------------|-------------------------|-----------------|
| Variables   |                         |                 |
| Constraints | Permutation Constraints | Marginals Fixed |
| Type        |                         |                 |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |

| Complexity | Cubic | Polynomial* |
|------------|-------|-------------|
|------------|-------|-------------|

|             | Monge                   | Kantorovich     |
|-------------|-------------------------|-----------------|
| Variables   |                         |                 |
| Constraints | Permutation Constraints | Marginals Fixed |
| Type        |                         |                 |
| Complexity  | Cubic                   | Polynomial*     |
|             |                         |                 |
|             |                         |                 |
|             |                         |                 |

|                                                  |  |  |
|--------------------------------------------------|--|--|
| <div data-bbox="405 181 2154 412">Solution</div> |  |  |
|--------------------------------------------------|--|--|

|             | Monge                   | Kantorovich     |
|-------------|-------------------------|-----------------|
| Variables   |                         |                 |
| Constraints | Permutation Constraints | Marginals Fixed |
| Type        |                         |                 |
| Complexity  | Cubic                   | Polynomial*     |
| Solution    |                         |                 |
|             |                         |                 |
|             |                         |                 |



|             | Monge                   | Kantorovich     |
|-------------|-------------------------|-----------------|
| Variables   |                         |                 |
| Constraints | Permutation Constraints | Marginals Fixed |
| Type        |                         |                 |
| Complexity  | Cubic                   | Polynomial*     |
| Solution    |                         |                 |
| Feasibility | Not always              | Always          |
|             |                         |                 |

|                                                                         |  |  |
|-------------------------------------------------------------------------|--|--|
| <div data-bbox="0 0 2453 588" data-label="Text"><p>Uniqueness</p></div> |  |  |
|-------------------------------------------------------------------------|--|--|

# Monge vs Kantorovich

|             | Monge                   | Kantorovich               |
|-------------|-------------------------|---------------------------|
| Variables   | $n$                     | $n^2$                     |
| Constraints | Permutation Constraints | Marginals Fixed           |
| Type        | Combinatorial           | Linear Program            |
| Complexity  | Cubic                   | Polynomial*               |
| Solution    | Permutation (matching)  | Coupling (transport plan) |
| Feasibility | Not always              | Always                    |
| Uniqueness  | Not always              | Not always                |

# From Discrete to Continuous Couplings

**Discrete**

**Continuous**